

Bilal Ahmed

Senior AI Engineer | Agentic AI | LLM & RAG Systems

Leeds, United Kingdom | +44 752 143 8761 | bilalah.dev07@gmail.com

LinkedIn: [linkedin.com/in/bilal-dev9](https://www.linkedin.com/in/bilal-dev9)

PROFESSIONAL SUMMARY

Senior AI Engineer with 6+ years of experience designing and deploying production-grade AI systems and scalable backend architectures. Specialised in Agentic AI and LLM-powered applications, including multi-agent orchestration (LangGraph, LangChain), RAG pipelines, tool/function calling, prompt engineering and vector search. Builds high-performance AI services with Python (FastAPI, Django) and Node.js, and deploys cloud-native systems on AWS, Google Cloud Platform (GCP) and Kubernetes. Proven track record of integrating OpenAI, Claude, Hugging Face, Llama and multimodal models into enterprise applications to automate workflows, improve user engagement and deliver measurable performance gains.

CORE COMPETENCIES

Agentic AI • Multi-Agent Systems • Agent Orchestration • Generative AI • Large Language Models (LLMs) • Retrieval-Augmented Generation (RAG) • Prompt Engineering • Tool & Function Calling • Model Context Protocol (MCP) • Vector Search & Embeddings • LLM Fine-Tuning • MLOps & LLMOps • Production AI Systems • Microservices & APIs • AWS • Google Cloud Platform (GCP) • Kubernetes • CI/CD • AI Workflow Automation

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, Bash

Agentic AI: Agentic Workflows, Multi-Agent Systems, Agent Orchestration, LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, Model Context Protocol (MCP), Tool Calling, Function Calling, Agent Memory & State Management, Planning & Reasoning (ReAct), Human-in-the-Loop, Agent Evaluation

Generative AI & LLMs: LLMs, RAG, Prompt Engineering, Context Engineering, Fine-Tuning, RLHF, Hugging Face Transformers, OpenAI APIs, Claude, Llama, Gemini, Multimodal Models, Structured Outputs, Guardrails, LLM Evaluation

Machine Learning & NLP: PyTorch, TensorFlow, NLP, OCR, ML Pipelines, MLflow, MLOps, Model Deployment & Monitoring

Vector Search & Data: Vector Databases (Pinecone, ChromaDB, FAISS, pgvector), Embeddings, Semantic Search, Hybrid Search, Reranking, Chunking Strategies

Backend Development: FastAPI, Node.js, Django, REST APIs, GraphQL, Microservices Architecture, WebSockets, Event-Driven Systems, Asynchronous Programming

Frontend Development: React, Next.js, TypeScript, TailwindCSS, Angular, Vue.js

Cloud & DevOps: AWS (EC2, S3, RDS, ECS, ECR, Lambda), Google Cloud Platform (Vertex AI, Cloud Run, GKE, BigQuery, Cloud Storage, Pub/Sub), Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins, Apache Kafka

Automation & AI Workflows: n8n, Zapier, Workflow Automation, AI Orchestration, Event-Based Systems

Databases: PostgreSQL, MongoDB, MySQL, Redis

Tools & Practices: Git, GitHub, Bitbucket, OpenAI APIs, Claude, Agile/Scrum, System Design, Code Review

PROFESSIONAL EXPERIENCE

Senior AI Engineer | The Codest

Nov 2023 – Present

- Designed and deployed agentic AI and LLM-powered applications for fintech platforms, using Retrieval-Augmented Generation (RAG) to improve the accuracy and contextual understanding of responses in financial workflows such as customer support, onboarding and compliance queries.
- Built scalable, secure AI microservices using Python and FastAPI to support real-time inference and high-throughput, low-latency fintech systems, deployed on AWS (ECS, ECR, S3, RDS) and Google Cloud Platform (Cloud Run, GKE).
- Integrated OpenAI, Hugging Face and Llama models into enterprise fintech platforms for conversational AI, tool-calling agents and workflow automation across financial operations.
- Implemented vector search systems using Pinecone and ChromaDB to enable semantic search and knowledge retrieval over financial documents, policies and customer data.
- Built multi-agent workflows with LangGraph and LangChain to automate document-heavy, multi-step financial processes, with human-in-the-loop review for sensitive decisions.
- Containerised services with Docker and orchestrated them on Kubernetes, with CI/CD pipelines (GitHub Actions) across AWS and GCP for reliable, repeatable releases.
- Collaborated with product and engineering teams to deliver AI-driven automation workflows that increased operational efficiency by 30%.

- Built full-stack applications integrating AI-powered backend systems with modern frontend frameworks, improving scalability and user experience.
- Developed LLM-driven features including intelligent search, conversational assistants and recommendation systems.
- Designed and maintained scalable APIs using Python (FastAPI) and Node.js within a microservices architecture.
- Implemented RAG pipelines to connect LLMs with enterprise data, enabling context-aware responses.
- Increased user engagement by ~30% through AI-enhanced search and recommendation capabilities.

- Developed full-stack web applications using Python (FastAPI) and React/TypeScript for scalable and maintainable systems.
- Built high-performance REST APIs supporting real-time data processing and complex business workflows.
- Architected backend services capable of handling high-volume data pipelines and concurrent user requests.
- Optimised PostgreSQL queries and indexing strategies, reducing query execution time by 25%.
- Deployed and maintained applications on AWS infrastructure to ensure reliable and scalable cloud services.

PROJECTS

Amelia – Enterprise Conversational AI Platform

Technologies: Python, FastAPI, RAG, LangGraph, Agentic AI, Vector Databases, AWS

- Built a conversational AI system for automated customer support and enterprise workflows.
- Implemented RAG-based knowledge retrieval for accurate, context-aware responses.
- Developed multi-agent orchestration pipelines for automation and decision support.

Design.ai – AI Content Generation Platform

Technologies: Python, Node.js, LangChain, LLM APIs, Multimodal AI, AWS

- Developed an AI platform capable of generating text, images and videos.
- Integrated multiple generative AI models for multimodal content creation.
- Built scalable backend services and cloud-based deployment architecture.

Sabermine – AI Document Processing System

Technologies: Python, FastAPI, OCR, NLP, AWS

- Built an AI system for automated document parsing and structured data extraction.
- Processed PDFs and scanned documents using NLP pipelines and OCR.
- Designed scalable processing pipelines for high-volume document workflows.

Myaigi – Multi-Agent AI System

Technologies: Node.js, JavaScript, LangChain, LangGraph, Agentic AI, LLMs, RAG

- Developed a multi-agent AI orchestration system for complex workflows.
- Built a modular agent architecture for task-specific AI execution with tool calling.
- Implemented reasoning pipelines using LLM-based decision systems.

SeekSocial – Social Learning Platform

Technologies: React, Node.js, PostgreSQL

- Built a collaborative learning platform, increasing user engagement by 25%.
- Implemented scalable backend services for real-time user interaction.
- Optimised APIs and queries, reducing response time by 20%.

CERTIFICATIONS

- AWS Certified Developer – Associate
- ChatGPT Prompt Engineering for Developers – DeepLearning.AI
- LLM Applications Development with LangChain and OpenAI

EDUCATION

Bachelor's Degree, Computer Science (Aug 2016 – Aug 2020)

University of Leeds, Leeds, United Kingdom